

Study on angle-only relative navigation for unmanned aerial vehicle formation in GPS-denied environment

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Zixuan Zheng¹, Sha Wang², Baichun Gong² , Erlong Su³ and Hefei Huang²

Abstract

To satisfy the autonomous relative navigation demand of unmanned aerial vehicle (UAV) formation in a GPS-denied environment, an angle-only relative navigation algorithm is proposed. First, the relative states in 3-dimension space are decoupled into relative motion along the horizontal plane and altitude by applying the relative azimuth between the north-east-down global frame and the body-fixed horizontal frame belonging to UAVs. Second, a relative dynamic and measurement model considering the measurement deviation of sensors are established. This solves the problem of colored noise in a majority of sensors used in practical engineering. Third, a consensus constraint model of relative states is established by introducing a geometric constraint among formation members. Additionally, a consensus-extended Kalman filter is designed. Finally, the angle-only relative navigation method is validated using standard Monte Carlo simulations. The simulation results show that all the states of the system are observable when the observability criterion is satisfied. The relative position accuracy achieved using gyros with a drift rate of 50°/h, accelerometers with 1 mg deviation, and optical cameras with 1° Gaussian white noise is within 10 m (3 σ). Additionally, the method proposed in this study can be used to estimate the sensor measurement deviations effectively. Furthermore, it can be potentially applied to improve the absolute navigation accuracy of UAVs.

Keywords

Unmanned aerial vehicle, consensus-extended Kalman filter, angle-only navigation, observability analysis

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Introduction

Unmanned aerial vehicle (UAV) formation, a set of aerial robots, that is, drones that work together to achieve a specific goal, has attracted increased attention in military and civilian applications in recent decades.^{1,2} Compared to a single UAV, distributed intelligent systems based on networked autonomous UAVs are expected to be more capable, flexible, and robust in many applications, including mapping, inspection, and detection.^{3–5} For example, in a flight mission, the failure of a single UAV directly lead to the failure of the mission, but a redundancy in the UAV formation design allows a backup UAV to complete the mission when a member of the formation fails. The flight mission of a UAV formation is relatively complex, and it is necessary to control the relative position and orientation of the formation members in the coordinated flight process so that the entire formation can maintain a certain geometric construction.⁶ However, the premise of completing formation control is precise relative navigation among the formation members.

Accurate relative navigation information can be obtained in a short time using the inertial navigation equipment (inertial measurement unit, IMU) carried by the UAV, but the relative navigation error diverges over time as the IMU's error drifts. Presently, most UAVs use IMU combined with GPS localization, which has been widely adopted and researched owing to the low-cost, high-rate output frequency, and high reliability. However, satellite navigation information is prone to interference and shading.⁷ Therefore, it is vital to find a relative navigation

¹ School of Astronautics, Northwestern Polytechnical University, China

² College of Astronautics, Nanjing University of Aeronautics and Astronautics, Nanjing, China

³ Science and Technology on Space Physics Laboratory, China Academy of Launch Vehicle Technology, Beijing, China

Corresponding author:

Baichun Gong, College of Astronautics, Nanjing University of Aeronautics and Astronautics, 29 Yudao Street, Qinhuai District, Nanjing, Jiangsu 210016, China.
Email: baichun.gong@nuaa.edu.cn

method that is completely autonomous and meets the accuracy requirements.

The range-only relative navigation algorithms for the UAV formation by using onboard data link range sensor, which does not need to consider the sensor pointing problem that exists in optical and radio measurements, have been studied a lot in the past decade. Bradley⁸ presented a range-only relative navigation algorithm in 3-dimension space and concluded that the system is locally weakly observable. Mohammed⁹ improved the observability of a range-only relative navigation system by installing multiple communication beacons on a single UAV. However, the baseline of the installed beacon was limited by the actual size of the UAV. Therefore, this scheme is more suitable for relative navigation under short-range conditions. Xiong¹⁰ introduced a UAV as a reference beacon in the formation. In studies,^{11–14} multiple fixed points were used as reference beacons to study the relative positioning method of UWB ranging. Helm et al.¹⁵ studied the relative positioning of UAVs using velocity measurement information and ultra-wideband range information of the camera optical flow method. However, the velocity measurement accuracy of the optical flow method is greatly affected by the environment and speed of the aircraft, and the speed measurement information is unavailable in some cases. Gong et al.¹⁶ presented a range-based collaborative relative navigation algorithm for fixed-wing UAV formation by integrated ranging sensor with other commonly used onboard sensors such as low-cost IMUs but without optical flow measurement. Martel¹⁷ proposed a unique four-degree-of-freedom relative posing and positioning scheme using the distance measurement information of at least six times the ultra-wideband and displacement information of each aircraft from each origin measured by V-SLAM.

However, in a complex electromagnetic denial environment, the method of relying on data link ranging to achieve relative navigation for the formation may still be unavailable, which is the same as GNSS. In contrast, relying on optical cameras to achieve relative navigation has become a vital development direction because of the passive nature, that is, there is no need to radiate electromagnetic waves and receive communication information, but it passively receives optical signals. However, angle-only relative navigation algorithms are often unobservable or weakly observable because of the lack of relative distance information between aircrafts. Wang et al.¹⁸ studied the relative navigation algorithm of dual line-of-sight vectors by introducing the relative distance information between two spacecraft and deduced the error propagation law for this method. Woffinden¹⁹ and Gong²⁰ found that the observable performance of an angle-only relative navigation system can also be improved by installing the camera away from the center of mass of the spacecraft under close-range conditions. Wang et al.²¹ studied a robust sigma point filtering algorithm. In this algorithm, when the zero-mean Gaussian

measurement noise is polluted, it can obtain higher relative navigation estimation accuracy compared to the extended Kalman filter.

The main contribution of this study is to develop an angle-only relative navigation method for UAV formation. Although this method is independent of GPS location, camera-based velocity measurement, and multi-antenna ranging, it can provide sufficiently accurate and stable navigation information for formation control by integrating angle-measuring cameras and other commonly used low-cost sensors, such as IMU, altimeters, magnetometers, and pitot tubes. Unlike in previous research, the sensor deviations estimated by this angle-only relative navigation method can also be used to improve the absolute navigation accuracy of the UAV.

This paper begins with the establishment of the relative dynamic model and measurement model in the second section. In the third section, the consensus constraint model for formation members is constructed, based on which the consensus-extended Kalman filter (CEKF) is designed. In After that, the observability analysis of the proposed angle-only relative navigation algorithm based on the Lie derivative is presented. Next, Monte Carlo simulations with sensor uncertainties are presented to verify the theoretical results. Finally, the conclusions are presented in the last section.

Problem statement and relative model

Problem statement

In this study, a tri-UAV formation flight mission in a GPS-denied environment is considered. Assuming that each UAV is capable of absolute measurements from IMU, altimeter, pitot tube, and magnetometer and relative measurement from optical camera, the key problem solved in this work is to integrate absolute and relative measurements to achieve an acceptable estimation of relative position and velocity among UAV formation.

Reference frame definition

The reference frames used in this study are shown in Figure 1. The global reference frame, denoted as n , considers the north-east-down (NED) geographic frame, which is assumed to be an inertial frame. The origin of the body-fixed frame, denoted as b , is located at the center of gravity of the UAV, and the directions of the three axes point front-right-down. The origin of the body-fixed horizontal frame, denoted as h , is also located at the center of gravity of the UAV, but the x and y axes are always in the horizontal plane, whereas the z -axis points toward the ground. The frame h is defined in the process of rotating the frame n to coincide with the frame b according to the 321 rotation sequence, and the Euler angle relative to the frame b is the azimuth φ (north-east is a positive direction). Euler angles relative to the frame n are the pitch angle θ and roll angle γ .

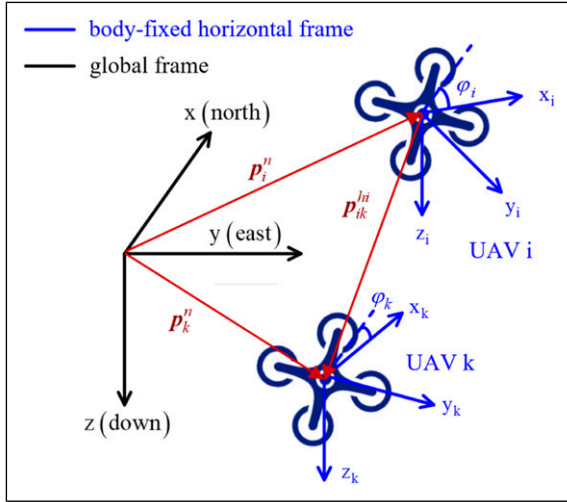


Figure 1. Illustration of the reference frames.

In contrast to the previous description of the relative motion between formation members in frame b , in this study, the relative dynamic model and measurement model concerning frame h are established. Here, the position vector of UAV $_i$ in frame n is denoted as $\mathbf{p}_i^n$, where the subscript i is the label of the UAV and the superscript n indicates that the vector is projected in frame n . Then, the projection of the position of UAV $_k$ relative to UAV $_i$ in the body-fixed horizontal frame h_i of the UAV $_i$ can be expressed as

$$\mathbf{p}_{ik}^{hi} = \mathbf{C}_n^{hi} (\mathbf{p}_k^n - \mathbf{p}_i^n) \quad (1)$$

where $\mathbf{C}_n^{hi}$ is the coordinate transformation matrix from n to h_i , which is related to the azimuth φ_i of the UAV $_i$

$$\mathbf{C}_n^{hi} = \begin{bmatrix} \cos\varphi_i & \sin\varphi_i & 0 \\ -\sin\varphi_i & \cos\varphi_i & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (2)$$

It can be observed from equations (1) and (2) that the z -axis of the relative position between UAVs is equal to the altitude difference in frame n , that is, the difference in the altitude of UAV $_i$ (which can be measured by an altimeter) and altitude of UAV $_k$. Therefore, the relative motion can be reduced from three dimensions in space to two dimensions in the horizontal plane.

Relative dynamic model

We establish the 15-dimension relative state vector between formation members, as shown in equation (3)^{15,16}

$$\mathbf{X}_{ik} = [\mathbf{p}_{ik}^{hi}, \psi_{ik}; \mathbf{v}_i^{hi}, \mathbf{v}_k^{hk}; \kappa_{oi}, \kappa_{ok}; \kappa_{ai}, \kappa_{ak}; v_i, v_k] \in \mathbb{R}^{15 \times 1} \quad (3)$$

where $\mathbf{p}_{ik}^{hi} \in \mathbb{R}^{2 \times 1}$ is the relative position of UAV $_k$ in the body-fixed horizontal frame of UAV $_i$. Vectors $\mathbf{v}_i^{hi}$ and $\mathbf{v}_k^{hk} \in \mathbb{R}^{2 \times 1}$ are the projections of the horizontal velocities of UAV $_i$ and UAV $_k$ in the frame h_i and h_k . Scalars κ_{oi} and κ_{ok} are the bias of the gyro measurements regarding the azimuth rates of UAV $_i$ and UAV $_k$. Vectors

κ_{ai} and $\kappa_{ak} \in \mathbb{R}^{2 \times 1}$ are the bias of the accelerometer measurements regarding the horizontal accelerations of UAV $_i$ and UAV $_k$. Scalars v_i and v_k are the bias of the horizontal speeds of UAV $_i$ and UAV $_k$ measured by pitot tubes. Finally, the relative azimuth ψ_{ik} is defined as

$$\psi_{ik} = \psi_k - \psi_i \quad (4)$$

We derive $\mathbf{X}_{ik}$ with respect to time t as follows

$$\mathbf{f}(\mathbf{X}_{ik}, \mathbf{U}_{ik}) = \frac{d\mathbf{X}_{ik}}{dt} = \begin{bmatrix} -\mathbf{v}_i^{hi} + \mathbf{C}_{hk}^{hi} \mathbf{v}_k^{hk} - (r_i - \kappa_{oi}) \mathbf{A} \mathbf{p}_{ik}^{hi} \\ (r_k - \kappa_{ok}) - (r_i - \kappa_{oi}) \\ (\mathbf{a}_i^{hi} - \kappa_{ai}) - (r_i - \kappa_{oi}) \mathbf{A} \mathbf{v}_i^{hi} \\ (\mathbf{a}_k^{hk} - \kappa_{ak}) - (r_i - \kappa_{oi}) \mathbf{A} \mathbf{v}_k^{hk} \\ 0_{8 \times 1} \end{bmatrix} \quad (5)$$

$$\mathbf{A} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (6)$$

where $\mathbf{C}_{hk}^{hi} \in \mathbb{R}^{2 \times 2}$ is a coordinate transformation matrix from frame h_k to frame h_i . System input is $\mathbf{U}_{ik} = [\mathbf{a}_i^{hi}, \mathbf{a}_k^{hk}, r_i, r_k] \in \mathbb{R}^{6 \times 1}$. Vector $\mathbf{a}_i^{hi} \in \mathbb{R}^{2 \times 1}$ is the horizontal acceleration of UAV $_i$ expressed in the frame h_i , which can be obtained from accelerometer measurements $\mathbf{s}_i \in \mathbb{R}^{3 \times 1}$. Scalar r_i is the azimuth rate obtained from gyro measurements $\boldsymbol{\omega}_i \in \mathbb{R}^{3 \times 1}$

$$\mathbf{a}_i^{hi} = \begin{bmatrix} \cos\theta_i & \sin\gamma_i \sin\theta_i & \cos\gamma_i \sin\theta_i \\ 0 & \cos\gamma_i & -\sin\gamma_i \end{bmatrix} \mathbf{s}_i \quad (7)$$

$$r_i = \frac{\sin\gamma_i}{\cos\theta_i} \omega_{ix} + \frac{\cos\gamma_i}{\cos\theta_i} \omega_{iz} \quad (8)$$

where ω_{ix} and ω_{iz} represent the x and z axes components of $\boldsymbol{\omega}_i$. For simplicity, the superscripts hi and hk for the state vector $\mathbf{X}_{ik}$ are omitted in the subsequent description.

Measurement model

In addition to the altimeter required in the subsection of problem statement, in this subsection, we establish measurement models for other sensors that are used to achieve relative navigation, including magnetometers, pitot tubes, cameras, and inertial measurement units. The magnetometer can be used to measure the azimuth of the UAV,²² and the measurement is modeled as

$$\tilde{\psi} = \psi + n(0, \varepsilon_\psi^2) \quad (9)$$

where ψ and $\tilde{\psi}$ represent the true and measurement values of the azimuth, respectively. The measurement noise was modeled as the Gaussian noise with zero mean and ε_ψ^2 variance. The pitot tube was used to measure the horizontal speed of the UAV, and the measurement was modeled as

$$\tilde{v} = v + v + n(0, \varepsilon_v^2) \quad (10)$$

where v and $\tilde{v}$ denote the true and measured values of the horizontal speed, respectively. The constant bias v of the

measurement error is primarily related to the local wind speed and UAV ground speed. The Gaussian white noise $n(0, \varepsilon_v^2)$ of the measurement error is mainly related to the measurement accuracy of the instrument itself.

The accuracy of relative measurements by optical camera will improve as the relative distance to the target decreases. Based on the imaging results of the target on the focal plane, the camera can obtain the relative line-of-sight angles, distance, attitude, and attitude angular velocity.²³ In this paper, we assume that the camera can only obtain the line-of-sight angles of the target, which brings less information but more challenges to the UAV formation relative navigation. Also, it is assumed that the measurement coordinate system coincides with the body-fixed frame b after the camera is installed, and the relative line-of-sight vector measured by the camera is projected onto the body-fixed horizontal frame h using the pitch angle and roll angle. Subsequently, the relative yaw angle between the two UAVs can be obtained. The camera measurement is modeled as follows

$$\tilde{\alpha} = \alpha + n(0, \varepsilon_\alpha^2) \quad (11)$$

where α and $\tilde{\alpha}$ represent the true and measurement values of the relative yaw, respectively, and ε_α^2 represents the Gaussian white noise variance. Alternatively, the camera measurement can also take the form of a line-of-sight vector

$$\tilde{l}_{ik} = \frac{p_{ik}^{hi}}{p_{ik}^{hi}} + n(0, \varepsilon_l^2) \quad (12)$$

where p_{ik}^{hi} is the norm of the vector p_{ik}^{hi} . Equations (11) and (12) are essentially equivalent, but the former will be used for subsequent simulations, whereas the latter will be used for the measurement model and observability analysis. The measurement errors of the accelerometers and gyroscopes are composed of two parts, constant deviation and Gaussian white noise, and the measurements are modeled as²⁴

$$\begin{aligned} \tilde{\omega} &= \omega + \kappa_\omega + n(0, \varepsilon_\omega^2) \\ \tilde{a} &= a + \kappa_a + n(0, \varepsilon_a^2) \\ \dot{\kappa}_\omega &= 0 \\ \dot{\kappa}_a &= 0 \end{aligned} \quad (13)$$

where $\tilde{\omega}$, ω , κ_ω , and $n(0, \varepsilon_\omega^2)$ represent the gyroscope measurements, true angular rates, constant bias of the gyro rates, and Gaussian white noise, respectively. $\tilde{a}$, a , κ_a , and $n(0, \varepsilon_a^2)$ represent accelerometer measurements, true accelerations, constant bias of the accelerometers, and Gaussian white noise, respectively.

According to the sensor measurement models established using equations (9), (10), (11), (12), and (13), the relative navigation system measurement equation can be obtained as follows

$$y = h(X_{ik}) = \begin{bmatrix} \tilde{l}_{ik} & \tilde{\psi}_{ik} & \tilde{v}_i & \tilde{v}_k \end{bmatrix} \in \mathbb{R}^{5 \times 1} \quad (14)$$

Design of distributed estimation filter

In this section, based on the extended Kalman filter, a CEKF is constructed by introducing the geometric topological relationship between the members of the UAV formation.

Consensus constraint model

Figure 2 illustrates the distributed sensor and communication network applied to multiple UAVs.

As shown in Figure 2, the relative position between every three UAVs satisfies

$$p_{ij} + C_j^i p_{jk} + C_k^i p_{ki} = 0 \quad (15)$$

Equation (15) is a physical constraint on the estimated state. In theory, the observability of the system increases when additional constraints are added to the system. Therefore, this equation constraint can be introduced into the filter to improve estimation performance. From equation (15), we can infer that

$$X_{ij} + T_j^i X_{jk} + T_k^i X_{ki} = 0 \quad (16)$$

where the generalized coordinate transformation matrixes T_j^i and T_k^i are defined as follows

$$T_j^i = \begin{bmatrix} C_{hj}^{hi} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -I_{2 \times 2} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & -I_{2 \times 2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & 0 \end{bmatrix} \in \mathbb{R}^{15 \times 15} \quad (17)$$

$$T_k^i = \begin{bmatrix} C_{hk}^{hi} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -I_{2 \times 2} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & -I_{2 \times 2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \in \mathbb{R}^{15 \times 15} \quad (18)$$

CEKF

The framework of the CEKF is illustrated in Figure 3. Based on the CEKF algorithm running on its hardware, all members estimate the relative states of their neighbors in the formation. Every CEKF integrates the measurements from onboard sensors together with the consensus constraint data from the other two UAVs. The formation members are required to share their data, including sensor measurements and prior estimation results of the CEKF. For example, to estimate the relative state of UAV₂ with respect to UAV₁ in the body-fixed horizontal frame of UAV₁, the measurement update is based on the prior estimation of UAV₂ and UAV₃. The time update equations for the CEKF²⁰ are given as

$$\frac{d\mathbf{X}}{dt} = \mathbf{f}(\mathbf{X}, \mathbf{U}) \quad (19)$$

$$\frac{d\mathbf{P}}{dt} = \mathbf{F}_X \mathbf{P} + \mathbf{P} \mathbf{F}_X^T + \mathbf{J}_U \mathbf{Q}_U \mathbf{J}_U^T \quad (20)$$

$$\mathbf{F}_X = \frac{\partial \mathbf{f}}{\partial \mathbf{X}} = \begin{bmatrix} -(r_i - \kappa_{oi})\mathbf{A} & \mathbf{A}\mathbf{C}_{hk}^{hi}\mathbf{v}_k^{hk} & -\mathbf{I}_{2 \times 2} & \mathbf{C}_{hk}^{hi} & \mathbf{p}_{ik}^{hi} & \mathbf{0}_{2 \times 1} & \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} \\ \mathbf{0}_{1 \times 2} & \mathbf{0} & \mathbf{0}_{1 \times 2} & \mathbf{0}_{1 \times 2} & 1 & -1 & \mathbf{0}_{1 \times 2} & \mathbf{0}_{1 \times 2} & \mathbf{0}_{1 \times 2} \\ \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 1} & -(r_i - \kappa_{oi})\mathbf{A} & \mathbf{0}_{2 \times 2} & \mathbf{A}\mathbf{v}_i^{hi} & \mathbf{0}_{2 \times 1} & -\mathbf{I}_{2 \times 2} & \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} \\ \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 1} & \mathbf{0}_{2 \times 2} & -(r_k - \kappa_{ok})\mathbf{A} & \mathbf{0}_{2 \times 1} & \mathbf{A}\mathbf{v}_j^{hj} & \mathbf{0}_{2 \times 2} & -\mathbf{I}_{2 \times 2} & \mathbf{0}_{2 \times 2} \\ \mathbf{0}_{8 \times 1} & \mathbf{0}_{8 \times 1} & \mathbf{0}_{8 \times 2} & \mathbf{0}_{8 \times 2} & \mathbf{0}_{8 \times 1} & \mathbf{0}_{8 \times 1} & \mathbf{0}_{8 \times 2} & \mathbf{0}_{8 \times 2} & \mathbf{0}_{8 \times 2} \end{bmatrix} \quad (21)$$

$$\mathbf{J}_U = \frac{\partial \mathbf{f}}{\partial \mathbf{U}} = \begin{bmatrix} \mathbf{0}_{2 \times 2} & \mathbf{0}_{2 \times 2} & -\mathbf{A}\mathbf{p}_{ik}^{hi} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{1 \times 2} & \mathbf{0}_{1 \times 2} & -1 & 1 \\ \mathbf{I}_{2 \times 2} & \mathbf{0}_{2 \times 2} & -\mathbf{A}\mathbf{v}_i^{hi} & \mathbf{0}_{2 \times 1} \\ \mathbf{0}_{2 \times 2} & \mathbf{I}_{2 \times 2} & \mathbf{0}_{2 \times 1} & -\mathbf{A}\mathbf{v}_j^{hj} \\ \mathbf{0}_{8 \times 6} \end{bmatrix} \quad (22)$$

The measurement update equations are given as^{25,26}

$$\mathbf{X}_t = \mathbf{X}_{t/t-1} + \mathbf{K}[\mathbf{z}_t - \mathbf{h}(\mathbf{X}_{t/t-1}, 0)] - \gamma \frac{\mathbf{P}_{t/t-1}}{1 + \|\mathbf{P}_{t/t-1}\|_F} (\mathbf{X}_{t/t-1} - \mathbf{X}_{con}) \quad (23)$$

$$\mathbf{P}_t = (\mathbf{I} - \mathbf{K}\mathbf{H})\mathbf{P}_{t/t-1} \quad (24)$$

where $\gamma > 0$ represents the feedback gain of the consensus constraint, $\|\cdot\|_F$ represents the Frobenius norm, and $\mathbf{X}_{con}$ represents the estimated state vector obtained from the consensus constraint equation (16).

Observability analysis

The concept of observability was originally introduced by Kalman to solve the problem of system-state

determination. If the initial state of the system can be uniquely determined by past observations, then the system is said to be observable. Observability reflects the ability of the filter to determine the state of the system using observations in a limited time and to a large extent determines whether the state vector can be estimated effectively.

Lie derivative

For the observability of nonlinear systems, a common practice is to analyze the observable rank condition of Lie derivatives. For a given nonlinear system

$$\Sigma: \begin{cases} \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}) \\ \mathbf{y} = \mathbf{h}(\mathbf{x}) \end{cases} \quad (25)$$

where $\mathbf{x} = [x_1, x_2, \dots, x_n]^T \in \mathbb{R}^{n \times 1}$ denotes the system state, $\mathbf{u} = [u_1, u_2, \dots, u_l]^T \in \mathbb{R}^{l \times 1}$ denotes the input, and $\mathbf{y} = [y_1, y_2, \dots, y_m]^T \in \mathbb{R}^{m \times 1}$ denotes the observation. Then

$$\mathbf{N}(\mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} \begin{bmatrix} L_f^0 \mathbf{h}(\mathbf{x}) \\ L_f^1 \mathbf{h}(\mathbf{x}) \\ \vdots \\ L_f^{n-1} \mathbf{h}(\mathbf{x}) \end{bmatrix} \in \mathbb{R}^{mn \times n} \quad (26)$$

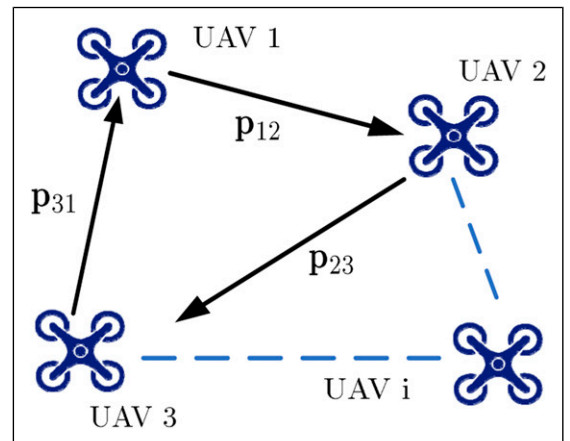


Figure 2. Measurement and communication topology diagram for UAV formation.

where

$$\begin{cases} L_f^0 \mathbf{h}(\mathbf{x}) = \mathbf{h} \in \mathbb{R}^{m \times 1} \\ L_f^i \mathbf{h}(\mathbf{x}) = \frac{\partial L_f^{i-1} \mathbf{h}(\mathbf{x})}{\partial \mathbf{x}} \mathbf{f}(\mathbf{x}) \in \mathbb{R}^{m \times 1} \end{cases} \quad (27)$$

If $\text{rank}(N) = n$, then the system is locally observable. According to the local weak observability theory of nonlinear systems, when the observability matrix composed of low-order Lie derivatives is not full of rank, high-order Lie derivatives must be introduced for observability until the observability matrix is full of rank. However, if higher order Lie derivatives still fail to satisfy the observability condition, the system is unobservable.

Observability analysis

Along with the deviations of the inertial measurement units and pitot tube, we must introduce high-order Lie derivatives to obtain observable conditions. However, consequently, the observability matrix acquires multiple dimensions, which make it difficult to solve. Note that the observability analysis of the system is difficult in theory. Hence, a numerical analysis is performed in Section 4.

Without considering the constant bias of the sensors, the state vector of the system will be reduced to 7 dimensions; however, the observation vector will maintain 5 dimensions. Therefore, two Lie derivatives are required to obtain potentially sufficient conditions for the system to be observable. After deduction, the observability matrix is obtained as follows

$$N = \begin{bmatrix} \mathbf{M}_1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \mathbf{v}_i^T & 0 \\ 0 & 0 & 0 & \mathbf{v}_k^T \\ \mathbf{M}_2 & \mathbf{M}_3 & \mathbf{M}_4 & \mathbf{M}_5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{a}_i^T & 0 \\ 0 & 0 & 0 & \mathbf{a}_k^T \end{bmatrix} \in \mathbb{R}^{10 \times 7} \quad (28)$$

$$\mathbf{M}_1 = \frac{1}{p_{ik}^3} (p_{ik}^2 \mathbf{I}_{2 \times 2} - \mathbf{p}_{ik} \mathbf{p}_{ik}^T) \quad (29)$$

$$\begin{aligned} \mathbf{M}_2 = & -\frac{r_i \mathbf{A}}{p_{ik}^3} (p_{ik}^2 \mathbf{I}_{2 \times 2} - \mathbf{p}_{ik} \mathbf{p}_{ik}^T) - \frac{\mathbf{v}_{ik} \mathbf{p}_{ik}^T}{p_{ik}^3} \\ & - \frac{(\mathbf{p}_{ik}^T \mathbf{v}_{ik}) \mathbf{I}_{2 \times 2} + \mathbf{p}_{ik} \mathbf{v}_{ik}^T}{p_{ik}^3} + \frac{3 \mathbf{p}_{ik} \mathbf{p}_{ik}^T \mathbf{v}_{ik} \mathbf{p}_{ik}^T}{p_{ik}^5} \end{aligned} \quad (30)$$

For equation (28), $\mathbf{M}_3$, $\mathbf{M}_4$, and $\mathbf{M}_5$ do not affect the observability analysis result and therefore not given. The necessary and sufficient condition for matrix N to be full rank is that the linear equations $N\mathbf{c} = 0$ concerning

$\mathbf{c} \in \mathbb{R}^{7 \times 1}$ have a unique solution $\mathbf{c} = 0$. After expansion, we obtain

$$\begin{cases} \mathbf{M}_1 \mathbf{c}_1 = 0 \\ c_2 = 0 \\ \mathbf{v}_i^T \mathbf{c}_3 = 0 \\ \mathbf{v}_k^T \mathbf{c}_4 = 0 \\ \mathbf{M}_2 \mathbf{c}_1 + \mathbf{M}_3 \mathbf{c}_2 + \mathbf{M}_4 \mathbf{c}_3 + \mathbf{M}_5 \mathbf{c}_4 = 0 \\ \mathbf{a}_i^T \mathbf{c}_3 = 0 \\ \mathbf{a}_k^T \mathbf{c}_4 = 0 \end{cases} \quad (31)$$

From the first sub-equation of equation (31), we obtain

$$\mathbf{c}_1 = \alpha \mathbf{p}_{ik} \quad (32)$$

If the velocity $\mathbf{v}_i$ and acceleration $\mathbf{a}_i$ of the UAV_{*i*} are not parallel, the 3rd, 4th, 6th, and 7th sub-equations of equation (31) indicate that

$$\mathbf{c}_3 = \mathbf{c}_4 = 0 \quad (33)$$

Substituting equations (32) and (33) into the 5th sub-equation of equation (31) yields

$$\mathbf{M}_1 \mathbf{c}_1 = -\alpha \mathbf{v}_{ik} \mathbf{p}_{ik}^T \mathbf{p}_{ik} + \alpha \mathbf{p}_{ik}^T \mathbf{v}_{ik} \mathbf{p}_{ik} = 0 \quad (34)$$

Let $\mathbf{p}_{ik} = [x; y]$, $\mathbf{v}_{ik} = [v_x; v_y]$, we obtain

$$\alpha (v_y x - v_x y) \begin{bmatrix} y \\ -x \end{bmatrix} = 0 \quad (35)$$

If $\mathbf{p}_{ik}$ is not parallel to $\mathbf{v}_{ik}$, it can be inferred that

$$v_y x - v_x y \neq 0 \quad (36)$$

From equations (35) and (36), we obtain $\alpha = 0$, which means $\mathbf{c}_1 = 0$. The aforementioned analysis shows that when the movement of the UAV formation satisfies $\mathbf{v}_i \nparallel \mathbf{a}_i$ and $\mathbf{p}_{ik} \nparallel \mathbf{v}_{ik}$, there is only a zero solution for $N\mathbf{c} = 0$, that is, the system can be observed. This raises the requirement for the absolute and relative movements of UAVs. First, the velocity and acceleration of a single UAV cannot have a parallel relationship. Second, the relative position and relative velocity between the formation members cannot be in a parallel relationship.

Simulation validation

The numerical simulation environment was established and the validity of the angle-only relative navigation algorithm was validated using the CEKF.

Trajectory design

The UAVs maintained a constant distance of 200 m in the horizontal plane, and the trajectory of the geometric center P of the formation is designed as (unit: m)

$$\mathbf{r}_P = \begin{bmatrix} 500 \\ 2500 \end{bmatrix} + \begin{bmatrix} 20t \\ 500 \sin(\omega_0 t) \end{bmatrix} \quad (37)$$

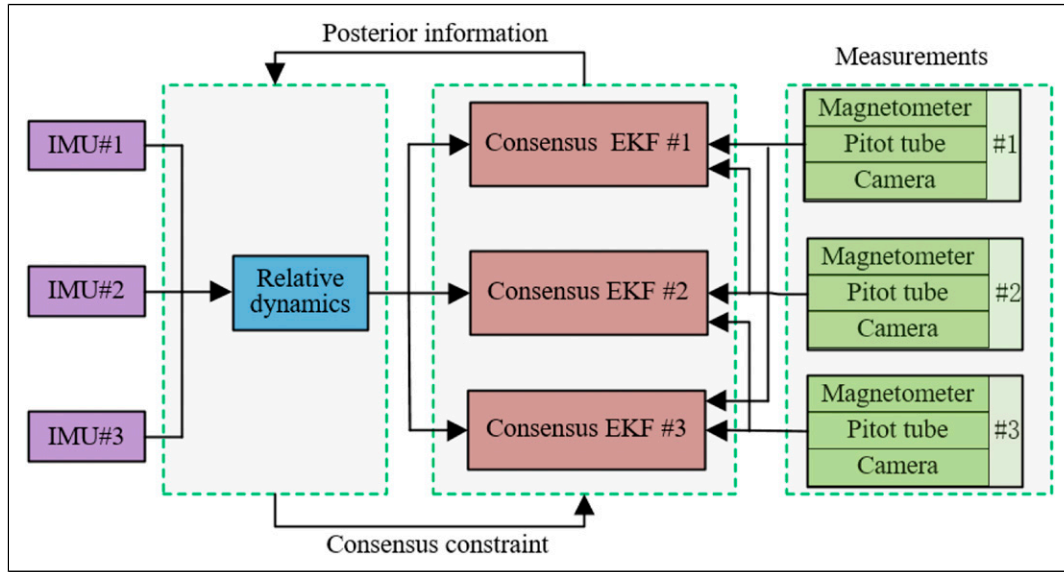


Figure 3. Framework of the consensus-extended Kalman filter.

where $\omega_0 = 0.030 \text{ rad/s}$. UAV₁, UAV₂, and UAV₃ perform circular motions with angular velocity $\omega = 0.015 \text{ rad/s}$ relative to P. Flying by the designed reference trajectories, the speed of the UAVs is between 18 m/s and 27 m/s, and the acceleration is less than 0.5 m/s^2 .

The angle between the velocity and acceleration of the UAVs within 300 s is shown in Figure 4. Because of the circular motion around P, the relative position and velocity are always vertical. Thus, the observability criterion discussed in Section 4.2 is satisfied and the system is observable. However, when $\omega = 0$, the relative velocity of the UAVs is 0. Thus, equation (23) does not hold and the system cannot be observed.

Simulation parameters

The sensor parameters and initial state estimation error parameter settings for the filter are presented in Tables 1 and 2, respectively. The constant bias measured by the gyroscope, accelerometer, and pitot tube was randomly generated in every Monte Carlo simulation. The filter frequency was 2 Hz and the total simulation time was 1800 s.

Simulation results

To test the performance of the proposed angle-only relative navigation algorithm, the following three groups of simulations were performed based on 200 Monte Carlo simulations. The EKF and CEKF algorithms were applied when the trajectories of the UAVs satisfy and differ from the observability criterion.

EKF applied to an unobservable system. By setting the consensus feedback gain $\gamma = 0$ for equation (23), the CEKF algorithm degenerates into the EKF algorithm. When $\omega = 0$, the system is unobservable. The state vector estimation error for the EKF algorithm is shown in Figure 5, Figure 6, Figure 7, and Figure 8.

Figure 5 shows the relative position estimation errors of the UAVs in the horizontal plane. The blue solid line is the mean error of 200 Monte Carlo simulations, whereas the yellow dashed line and red dash-dotted line are the $+3$ sigma and -3 sigma boundaries, respectively. As shown in Figure 5, the relative position estimation results were divergent. Additionally, the constant drift estimation results for the gyro and constant bias estimation results for the accelerometer along the x axis, as shown in Figure 7, are also divergent. However, some variables of the system are still observable. For velocity, as shown in Figure 4, the constant bias of the accelerometer along the y axis, as shown in Figure 5, the constant bias of the Pitot tube, and the relative azimuth, as shown in Figure 6, the estimation results remain convergent.

EKF applied to an observable system. When $\omega = 0.015 \text{ rad/s}$, the system is proven to be observable, and the estimation error of the EKF algorithm is shown in Figure 9, Figure 10, Figure 11, and Figure 12. After $t = 600$ s, the ± 3 sigma boundaries were close to the mean error, which oscillated around 0, indicating that the filter algorithm was convergent.

Figure 9 shows the relative position estimation errors of the UAVs in the horizontal plane. After the filter converges, the error boundaries of the x axis are between -8 m and $+8$ m, and the error boundaries of the y axis are between -6 m and $+6$ m. Thus, the relative position estimation accuracy of UAVs is within 10 m.

Figure 10 shows the velocity estimation errors of the UAVs. After the filter converged, the velocity estimation errors along the x and y axes were both within 0.2 m/s, indicating that the velocity estimation accuracy along the x axis was slightly better than that along the y axis.

Figure 11 shows the estimation error of the constant bias of the inertial measurement units. The three sub-figures on the left show the estimation results of the

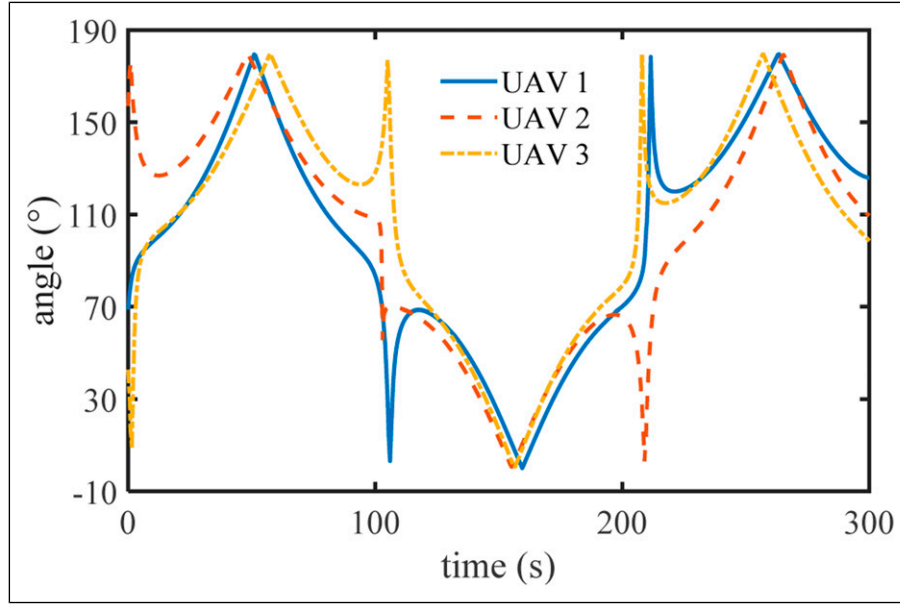


Figure 4. Angle between velocity and acceleration of UAVs.

Table 1. Parameter settings for sensors.

Parameter	Value	Parameter	Value
Gyro rate uncertainty	Bias 50°/h Noise 5°/h	Accelerometer uncertainty	Bias 1 mg Noise 0.1 mg
Azimuth noise	5°	Speed uncertainty	Bias 10 m/s Noise 1 m/s
Relative yaw noise	1°		

Table 2. Initial state estimation error parameter setting for the filter.

Parameter	Value	Parameter	Value
Relative position	1 m	Velocity	1 m/s
Relative azimuth	5°	Gyro bias	50°/h
Accelerometer bias	1 mg	Pitot tube bias	3 m/s

constant drift of the gyro, and the six sub-figures on the right show the estimation results of the constant bias of the accelerometer. The estimation error decreased with time, gradually approaching zero. For the constant drift estimation error of the gyro at 600 s, 1200 s, and 1800 s, it is less than 50°/h, 30°/h, and 20°/h, respectively. For the constant bias estimation error of the accelerometer along the x axis, it is less than 0.7 mg at 600 s and less than 0.5 mg after 1200 s. For the y axis, it is close to 0.01 mg after the filter converges.

Figure 12 shows the estimation error of the pitot tube speed bias and the relative azimuth. At 1800 s, the estimation accuracy of the pitot tube constant bias is approximately 0.14 m/s, and the relative azimuth is approximately 0.2°.

CEKF applied to an observable system. In theory, if additional constraints, such as increasing the number of

observations, are added to the system, the observability of the system will improve. To visually verify that the CEKF algorithm proposed in Section 4.2, which introduces consensus feedback in the measurement update equations, is better than the EKF algorithm, we set the consensus coefficient to be $\gamma = 0.03$ and defined the following statistics

$$e_m(k, t_0, t_1) = \frac{1}{t_1 - t_0} \int_{t_0}^{t_1} |\hat{X}_{ik}(k, t) - X_{ik}(k, t)| dt \quad (38)$$

where scalar e_m represents the average absolute difference between the estimated state $\hat{X}_{ik}$ and true state X_{ik} from the filter convergence time t_0 to the filter ending time t_1 in the k^{th} Monte Carlo simulation

$$M_m(k, t_0, t_1) = \frac{1}{n} \sum_{k=1}^n e_m \quad (39)$$

$$P_m(k, t_0, t_1) = \frac{1}{n-1} \sum_{k=1}^n [e_m - M_m][e_m - M_m]^T \quad (40)$$

where M_m and P_m are the mean and covariance errors, respectively, with which the relative navigation algorithm can be evaluated. Small M_m and P_m values usually indicate high estimation accuracy and good stability. Considering the filter convergence time as $t_0 = 600$ s, Table 3 lists the statistical results of the estimation error for

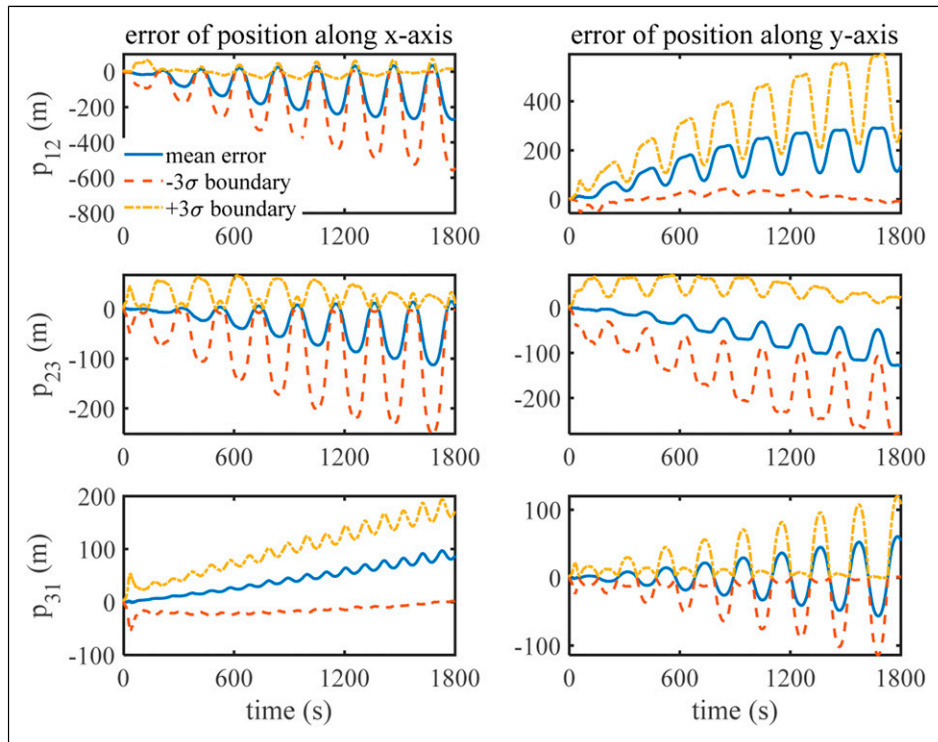


Figure 5. Estimation error of the relative position when $\omega = 0$.

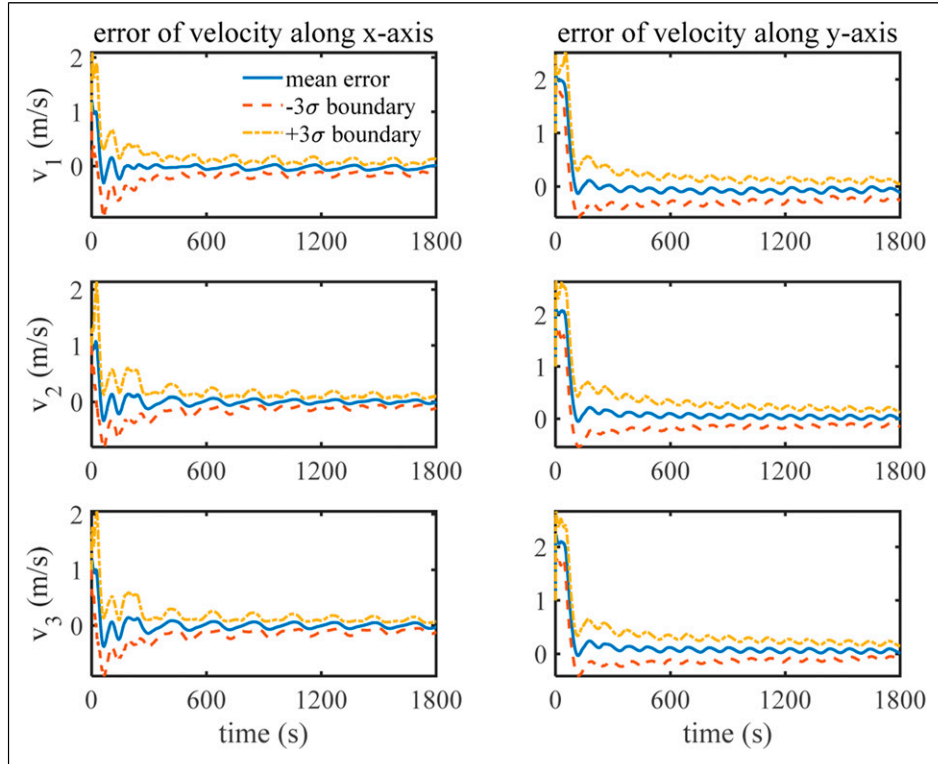


Figure 6. Estimation error for velocity when $\omega = 0$.

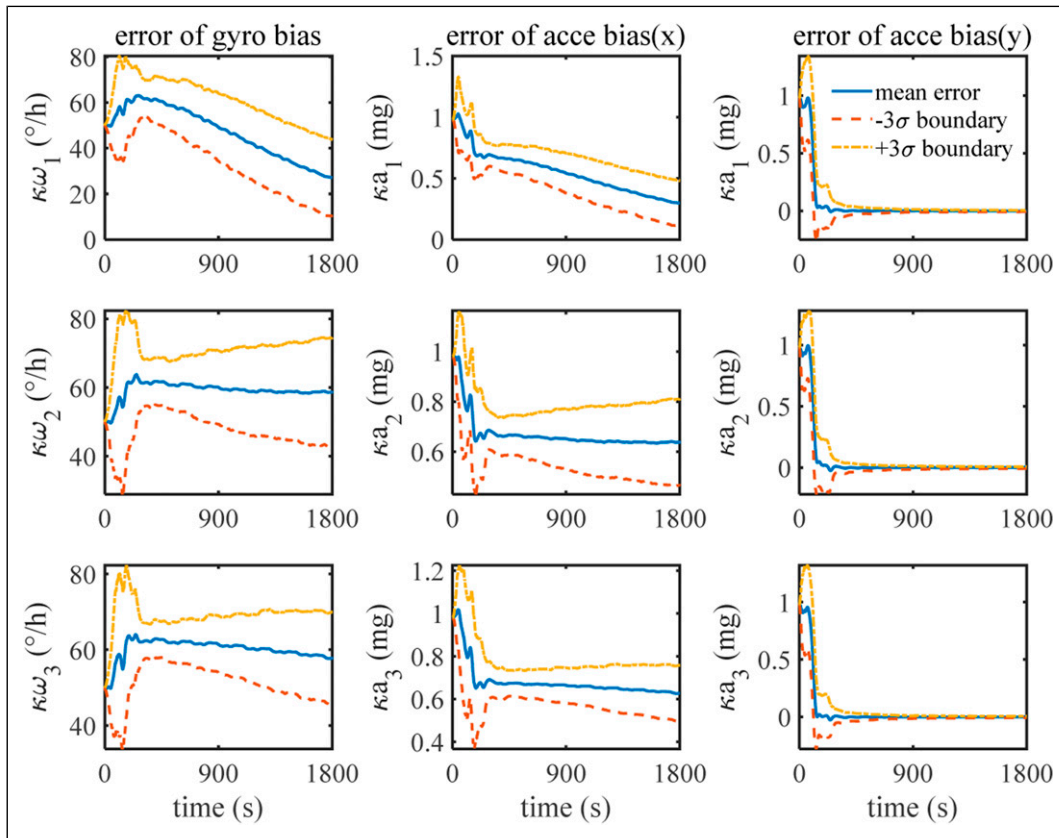


Figure 7. Estimation error for the inertial measurement unit bias when $\omega = 0$.

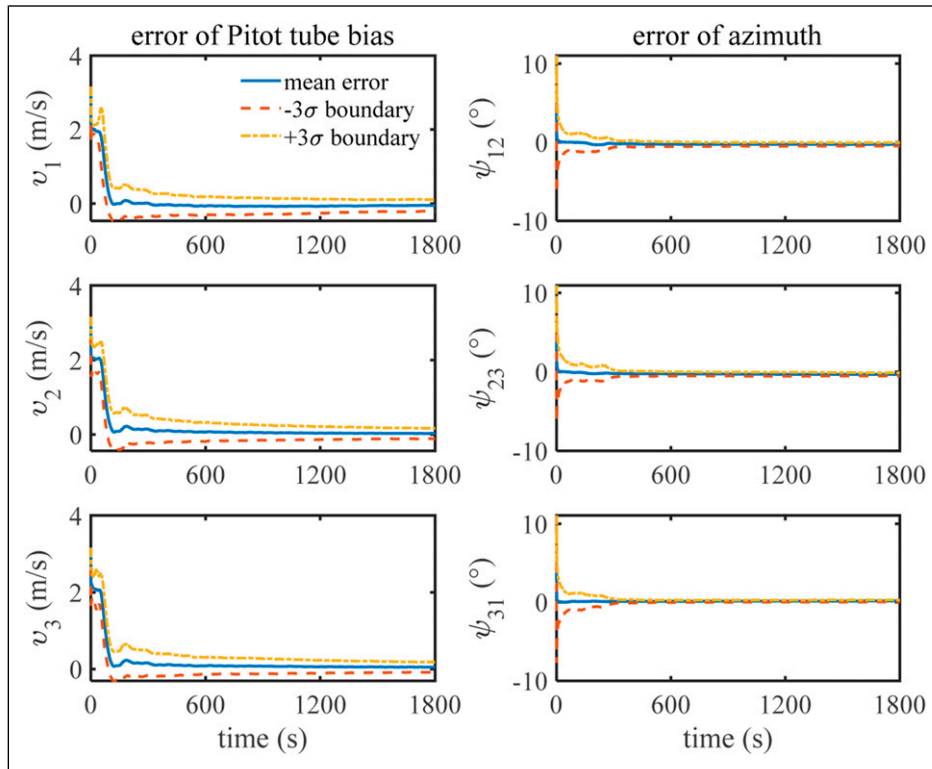


Figure 8. Estimation error for pitot tube bias and relative azimuth when $\omega = 0$.

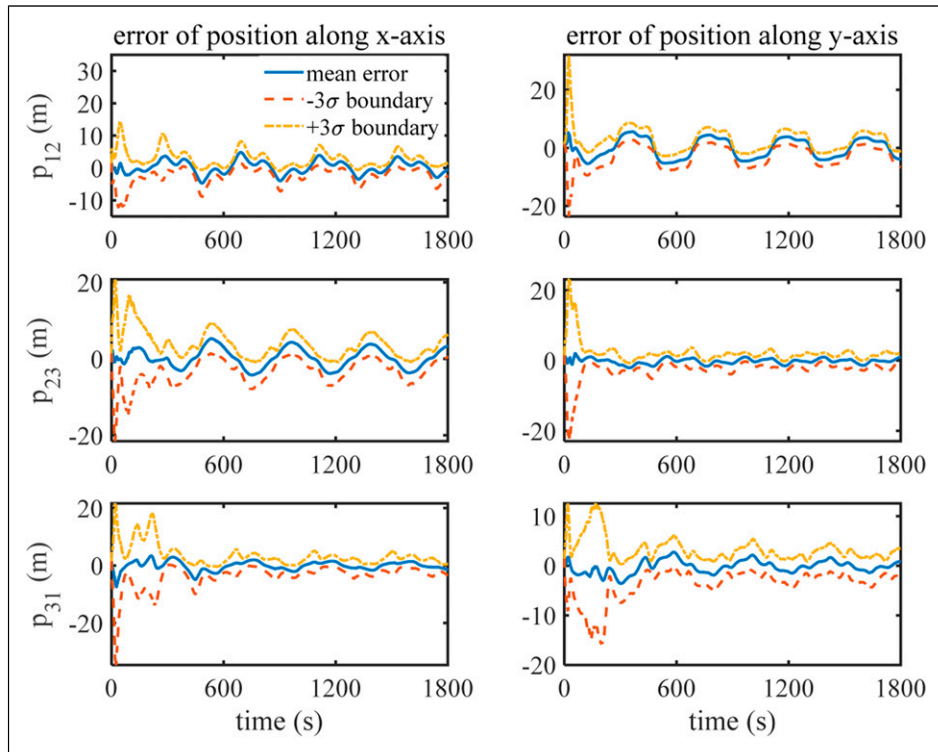


Figure 9. Estimation error for the relative position when $\omega = 0.015/\text{s}$.

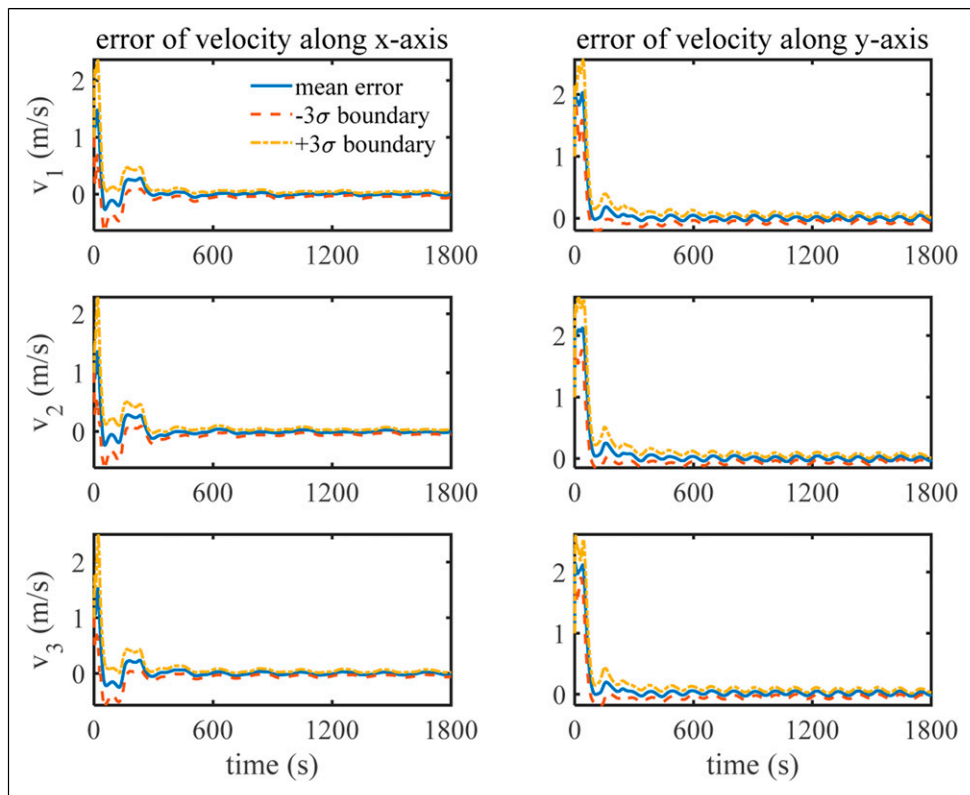


Figure 10. Estimation error for velocity when $\omega = 0.015/\text{s}$.

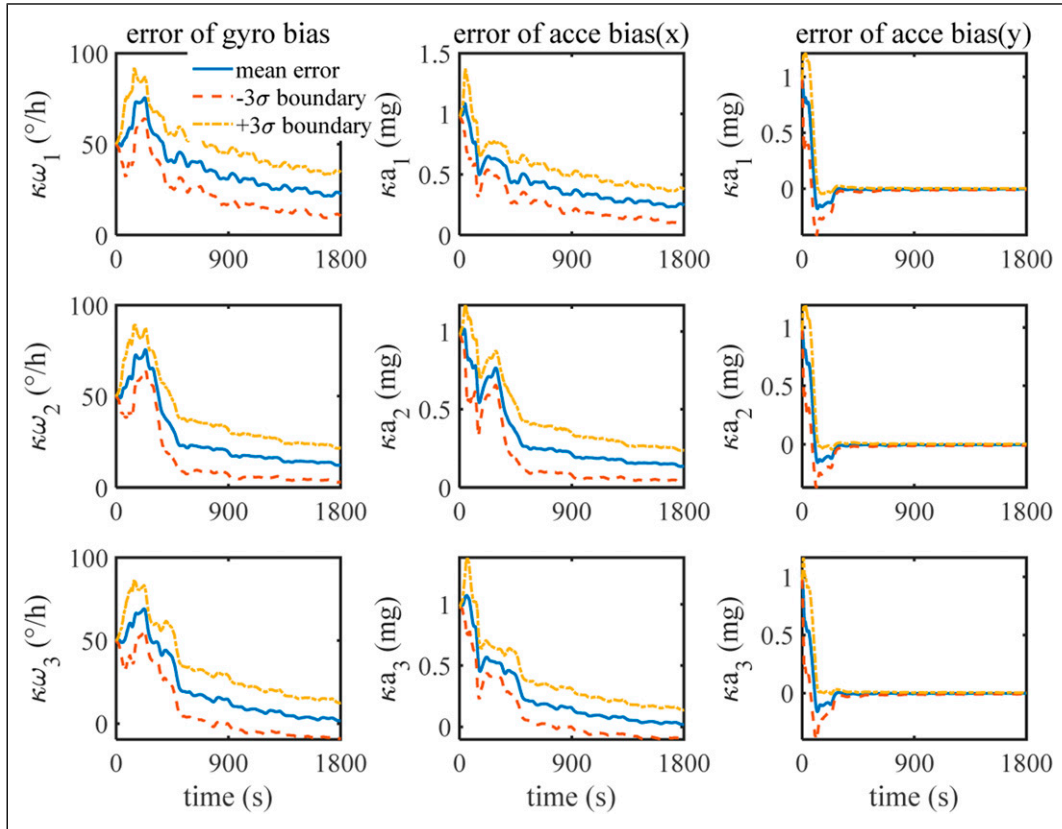


Figure 11. Estimation error for inertial measurement units bias when $\omega = 0.015/s$.

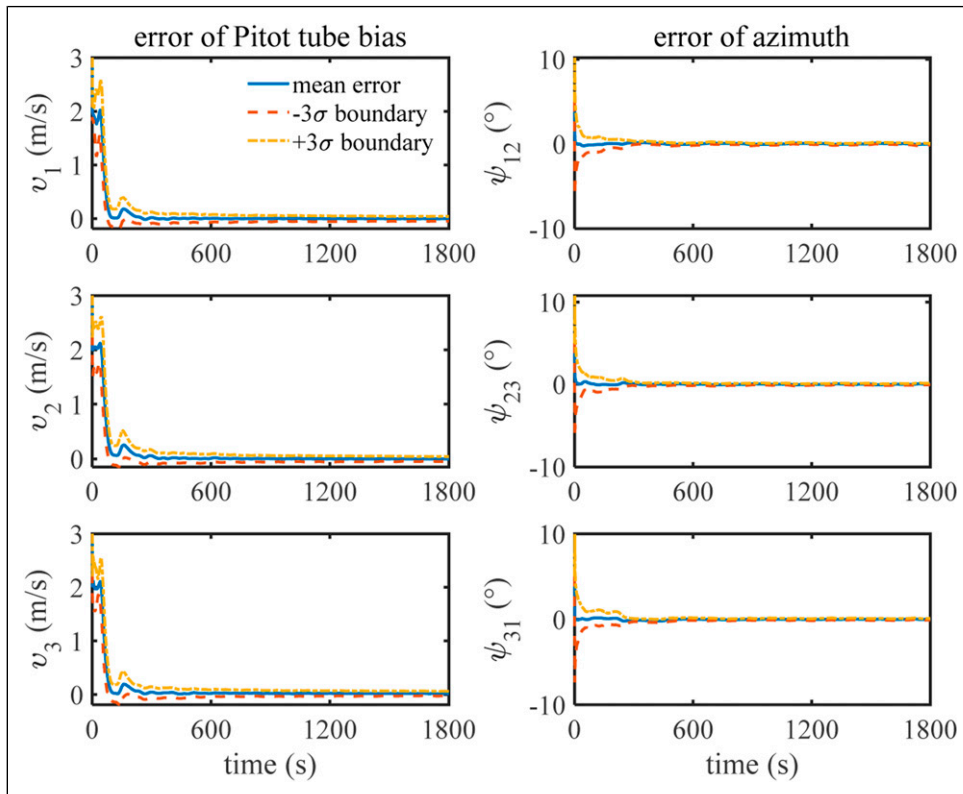


Figure 12. Estimation error for pitot tube bias and relative azimuth when $\omega = 0.015/s$.

Table 3. Statistical results of estimation error for EKF and CEKF algorithms.

		Mean error		Covariance error	
		EKF	CEKF	EKF	CEKF
Relative position (m)	x axis	1.718	1.382	1.169	0.833
	y axis	1.576	1.245	1.253	0.774
Velocity (m/s)	x axis	0.018	0.018	0.012	0.012
	y axis	0.028	0.028	0.018	0.017
Relative azimuth (°)		0.042	0.039	0.030	0.028
Gyro bias (°/h)		18.16	14.33	9.041	5.079
Accelerometer (mg)	x axis	0.203	0.160	0.100	0.056
	y axis	0.004	0.004	0.002	0.002
Pitot tube bias (m/s)		0.017	0.016	0.012	0.012

the EKF and CEKF algorithms when the system is observable. Compared with the EKF algorithm, the CEKF algorithm shows a smaller estimation mean error and covariance error, which validates that the CEKF algorithm designed in this study possesses a higher estimation accuracy and better stability.

Conclusion

A new angle-only cooperative relative navigation algorithm for UAV formation, which fuses an optical camera with other commonly used low-cost sensors, such as inertial measurement units, altimeters, magnetometers, and pitot tubes, was analyzed in this study. Accurate relative navigation for a formation composed of at least three UAVs was achieved with relatively light equipment, low power consumption, and low processing cost in a GPS-denied environment. The research work of this dissertation includes four main aspects.

- (1) An accurate relative dynamic model and measurement model between UAVs in the horizontal plane were established, and sensor measurement deviations were considered to realize the whitening of non-white Gaussian noise.
- (2) The Lie derivative criterion was introduced to analyze the observability of the angle-only relative navigation system, and sufficient conditions for the system to be observable were obtained. First, the velocity and acceleration of a single UAV cannot be parallel to each other. Second, the relative position and velocity between UAVs cannot be parallel.
- (3) A decentralized estimation strategy based on the CEKF was designed, and a consensus algorithm was constructed using multiple physical constraints.
- (4) The validity of the angle-only relative navigation algorithm and the superiority of the consensus-extended Kalman filter were validated using standard Monte Carlo simulations. All the states are

observable when the trajectories of the UAVs satisfy the observability criterion.

The angle-only relative navigation algorithm proposed in this study was based on three UAVs. When more UAVs will participate in the formation, the decentralized strategy and nonlinear estimation algorithm will become more complicated. This will be the focus of future research.

Declaration of conflicting interests

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ORCID iD

Baichun Gong  <https://orcid.org/0000-0002-6701-797X>

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